

state Vector:

$|0\rangle$ _____

$$|\psi\rangle = \begin{bmatrix} a \\ b \end{bmatrix}$$

$$= \begin{bmatrix} a \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ b \end{bmatrix}$$

$$= a \begin{bmatrix} 1 \\ 0 \end{bmatrix} + b \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$= a|0\rangle + b|1\rangle$$

$$|a|^2 + |b|^2 = 1$$

$p(0) \quad p(1)$

Unitary matrix:

$$U_{n \times n} = \begin{bmatrix} & \end{bmatrix}$$

$$U^\dagger = U^{-1}$$

$$U^\dagger U = I$$

$$U U^\dagger = I$$

(Ensure reversibility)

Single qubit gates:

Logical gates:

NOT gate

I	O
0	1
1	0

$0 \xrightarrow{\text{NOT}} 1 \xrightarrow{\text{NOT}} 0$
(Reversible)

Reset zero

I	O
0	0
1	0

$0 \xrightarrow{R} 0 \xrightarrow{R} 0$

$1 \xrightarrow{R} 0 \xrightarrow{R} 0$ (Not Reversible)

In quantum gates it should be reversible "each input should have a distinct output".

All closed-system quantum evolutions conserve probability and are described by unitary operators. $U^{-1} = U^\dagger$

pauli gates:

$$\sigma_x = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$\sigma_y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$$

$$\sigma_z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$\sigma_x |0\rangle = |1\rangle$$

* Matrix Mul

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}_{2 \times 2} \begin{bmatrix} 1 \\ 0 \end{bmatrix}_{2 \times 1} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}_{2 \times 1}$$

$$\sigma_z |0\rangle = |0\rangle$$

$$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\sigma_z |1\rangle = -|1\rangle$$

σ is Unitary: $\sigma^\dagger \sigma = I$

σ is Hermitian: $\sigma^\dagger = \sigma$

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$X X |0\rangle = X |1\rangle = |0\rangle$$

$$|\psi\rangle = a|0\rangle + b|1\rangle = \begin{bmatrix} a \\ b \end{bmatrix}$$

$$X |\psi\rangle = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} b \\ a \end{bmatrix}$$

$$|\psi'\rangle = b|0\rangle + a|1\rangle$$

$$\begin{aligned} X |\psi\rangle &= a X |0\rangle + b X |1\rangle \\ &= a |1\rangle + b |0\rangle \end{aligned}$$

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$

$$= |1\rangle\langle 0| + |0\rangle\langle 1|$$

↗ outer product

$$\begin{bmatrix} 0 \\ 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$$

$$\begin{aligned} X |0\rangle &= (|1\rangle\langle 0| + |0\rangle\langle 1|) |0\rangle \\ &= |1\rangle\langle 0|0\rangle + |0\rangle\langle 1|0\rangle \\ &= |1\rangle + 0 = \underline{|1\rangle} \end{aligned}$$

outer product:

$$\begin{array}{c} \text{output} \quad \text{input} \\ \downarrow \quad \downarrow \\ | \rangle \langle | \end{array}$$

$$Z = |0\rangle\langle 0| - |1\rangle\langle 1|$$

$$= \begin{bmatrix} 1 \\ 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \end{bmatrix} - \begin{bmatrix} 0 \\ 1 \end{bmatrix} \begin{bmatrix} 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} - \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = Z$$

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$H|0\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$H|0\rangle = \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle) = \frac{|0\rangle + |1\rangle}{\sqrt{2}} = |+\rangle$$

$$H|1\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}} = |-\rangle$$

$$H = |+\rangle\langle 0| + |-\rangle\langle 1|$$

$$|+\rangle = \frac{1}{\sqrt{2}} |0\rangle + \frac{1}{\sqrt{2}} |1\rangle$$

$$\rho(0) = \frac{1}{2}$$

$$\rho(1) = \frac{1}{2}$$

Super position

$$HH|\psi\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix}$$

$$= \frac{1}{2} \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = |\psi\rangle$$

Rotation gates:

Rotate by angle θ on pauli axis

$R_x(\theta)$ $R_y(\theta)$ $R_z(\theta)$

$$e^{\frac{-i\theta\sigma}{2}}$$

Pauli

$$I \cos\left(\frac{\theta}{2}\right) - i \sigma \sin\left(\frac{\theta}{2}\right)$$

for $\sigma_x \Rightarrow R_x$

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \cos\left(\frac{\theta}{2}\right) - i \sin\left(\frac{\theta}{2}\right) \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} \cos\left(\frac{\theta}{2}\right) & -i \sin\left(\frac{\theta}{2}\right) \\ -i \sin\left(\frac{\theta}{2}\right) & \cos\left(\frac{\theta}{2}\right) \end{bmatrix}$$

for $\theta = \pi$

$$R_x(\pi) = \begin{bmatrix} \cos\left(\frac{\pi}{2}\right) & -i \sin\left(\frac{\pi}{2}\right) \\ -i \sin\left(\frac{\pi}{2}\right) & \cos\left(\frac{\pi}{2}\right) \end{bmatrix}$$

$$= \begin{bmatrix} 0 & -i \\ -i & 0 \end{bmatrix} = -i \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

for σ_y

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \cos\left(\frac{\theta}{2}\right) - i \sin\left(\frac{\theta}{2}\right) \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$$

$$R_y(\theta) = \begin{bmatrix} \cos\left(\frac{\theta}{2}\right) & -\sin\left(\frac{\theta}{2}\right) \\ \sin\left(\frac{\theta}{2}\right) & \cos\left(\frac{\theta}{2}\right) \end{bmatrix}$$

for $\theta = \pi$ $R_y(\pi) = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$

for σ_z

$$R_z = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \cos\left(\frac{\theta}{2}\right) - i \sin\left(\frac{\theta}{2}\right) \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$\begin{bmatrix} \cos\left(\frac{\theta}{2}\right) - i \sin\left(\frac{\theta}{2}\right) & 0 \\ 0 & \cos\left(\frac{\theta}{2}\right) + i \sin\left(\frac{\theta}{2}\right) \end{bmatrix}$$

$$\cos(x) - i \sin(x) = e^{-ix}$$

$$\cos(x) + i \sin(x) = e^{ix} \quad \text{Euler Formula}$$

$$R_z(\theta) = \begin{bmatrix} e^{-i\frac{\theta}{2}} & 0 \\ 0 & e^{i\frac{\theta}{2}} \end{bmatrix}$$

for $\theta = \pi$

$$R_z(\pi) = \begin{bmatrix} e^{-i\frac{\pi}{2}} & 0 \\ 0 & e^{i\frac{\pi}{2}} \end{bmatrix}$$

$$= \begin{bmatrix} -i & 0 \\ 0 & i \end{bmatrix}$$

Any gate can be created by Rotation gates

$$U = e^{i\alpha} R_z(\beta) R_y(\lambda) R_z(\delta)$$

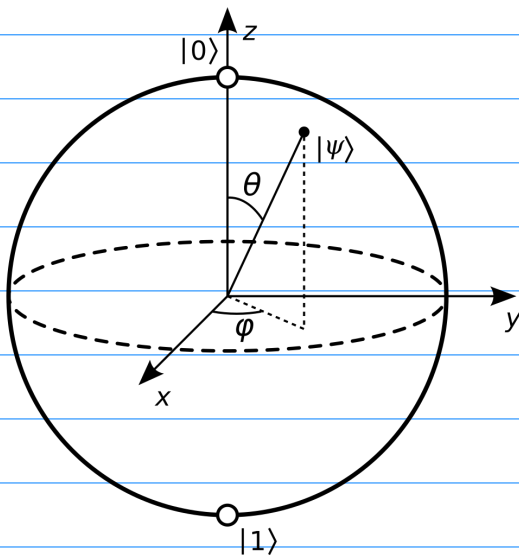
Euler decomposition of $SU(2)$

For $\alpha = \frac{\pi}{2}$ $\beta = 0$ $\gamma = \frac{\pi}{2}$ $\delta = \pi$

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

Bloch sphere: pure one qubit state
[one point on sphere]

$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right)|0\rangle + e^{i\phi}\sin\left(\frac{\theta}{2}\right)|1\rangle$$



$$r_x = \sin(\theta) \cos(\phi)$$

$$r_y = \sin(\theta) \sin(\phi)$$

$$r_z = \cos(\theta)$$

$$(r_x, r_y, r_z)$$

$$r_x^2 + r_y^2 + r_z^2$$

$$\sin^2(\theta) \cos^2(\phi) + \sin^2(\theta) \sin^2(\phi) + \cos^2(\theta)$$

$$= \sin^2(\theta) (\cos^2(\phi) + \sin^2(\phi)) + \cos^2(\theta)$$

$$= \sin^2(\theta) + \cos^2(\theta) = 1$$

$$(1, 0, 0)$$

$$r_x = 1$$

$$r_y = 0$$

$$r_z = 0$$

$$\theta = \frac{\pi}{2}$$

$$\phi = 0$$

$$\begin{aligned} |\psi\rangle &= \cos\left(\frac{\pi}{4}\right)|0\rangle + e^{i\phi}\sin\left(\frac{\pi}{4}\right)|1\rangle \\ &= \frac{|0\rangle + |1\rangle}{\sqrt{2}} = |+\rangle \quad \text{X axis pos} \end{aligned}$$

For Z axis pos neg
 $|0\rangle, |1\rangle$

For X axis $|+\rangle, |-\rangle$

For Y axis $|+i\rangle, |-i\rangle$

$$\frac{|0\rangle + i|1\rangle}{\sqrt{2}}, \quad \frac{|0\rangle - i|1\rangle}{\sqrt{2}}$$

Applying single gates:

$$|0\rangle \xrightarrow{\boxed{X}} |\psi\rangle$$

$$X|0\rangle = |1\rangle$$



$$H|0\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}} = |+\rangle$$

$$= |\psi_0\rangle$$

$$Z|\psi_0\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}} = |-\rangle$$

$$Z H |0\rangle =$$

$$\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} |0\rangle =$$

$$\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} |0\rangle =$$

$$\frac{|0\rangle - |1\rangle}{\sqrt{2}} = |-\rangle$$

Additional gates:

phase gate

$$ph(\theta) = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\theta} \end{bmatrix} \text{ General } Z$$

$$ph(\pi) = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = Z$$

$$ph(\theta)|\psi\rangle = a|0\rangle + e^{i\theta}b|1\rangle$$

$$\Theta = \frac{\pi}{2}$$

$$\rho h\left(\frac{\pi}{2}\right) = \begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix} = S \text{ gate}$$

$$S^2 = Z$$

$$S^\dagger = S^{-1} = S^3$$

$$\text{For } \Theta = \frac{\pi}{4}$$

$$\rho h\left(\frac{\pi}{4}\right) = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\frac{\pi}{4}} \end{bmatrix} = T \text{ gate}$$

$$T^2 = S$$

$$T^4 = Z$$

$H Z X Y |0\rangle :$

$H|T\rangle:$